Chapter 4 Z-Transform and its Application

→ Define

① Linear system:- A discrete-Time system is said to be Linear if it obeys the principles of superposition

$a_1 x_1(n) \longrightarrow a_1 y_1(n)$

$a_2 x_2(n) \longrightarrow a_2 y_2(n)$

$a_1 x_1(n) + a_2 x_2(n) \longrightarrow a_1 y_1(n) + a_2 y_2(n)$

② Time invariant system:- A discrete-Time system said to be Time invariant if its output is independent on the Time when the input is applied

$x(n) \longrightarrow y(n)$ & $x(n-1) \longrightarrow y(n-1)$

③ Causal system:- the system is used to be Causal system if it Produces an o/p only where there is an input

"No output befor Time Zero"

$h(k) = 0 \qquad k < 0$

④ Stability:- An LTIS is stable if its impulse respone $h(k)$, satisfied the following Condition

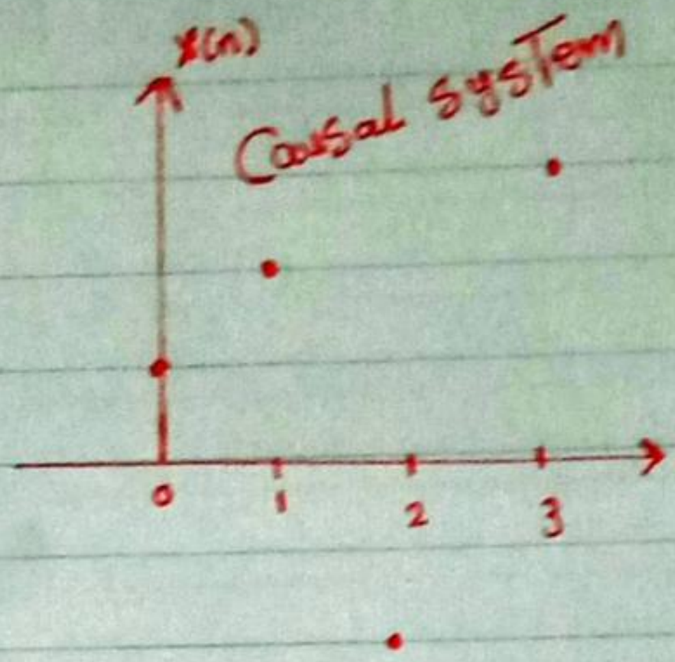
$$\sum_{k=-\infty}^{\infty} h(k) < \infty$$

⑤ Region of Convergence (ROC):- all values at which $X(z)$ is Finite ($X(z) \neq \infty$)

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* Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$



→ $x(n) = \{1, 3, -4, 5\}$

① Find $X(z)$

$$X(z) = \sum_{n=0}^{3} x(n) z^{-n}$$

$$= x(0)z^{-0} + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3}$$

$$= 1 + 3z^{-1} - 4z^{-2} + 5z^{-3}$$

$$X(z) = 1 + \frac{3}{z} - \frac{4}{z^2} + \frac{5}{z3} = \frac{z^3 + 3z^2 - 4z + 5}{z3}$$

* RoC = All Values of Z - { 0 }

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* The inverse - Z - Transform

$$X(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{b_0 + b_1 z^{-1} + b_2 z^{-2}}$$

IZT → Power series → Long Division
IZT → Power series → Recusively
IZT → Partial Fraction
IZT → Respdue (x)

① Power series at Long Division

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} + 0.3561 z^{-2}}$$

* Find x(n) 5 First Samples using Power series Long Division

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$$1+3z^{-1}+3.6439z^{-2}+2.57z^{-3}$$

$1-z^{-1}+0.3561z^{-2}$) $1+2z^{-1}+z^{-2}$

$1-z^{-1}+0.3561z^{-2}$

$3z^{-1}+0.6439z^{-2}$

$3z^{-1}-3z^{-2}+1.068z^{-3}$

$3.6439z^{-2}-1.068z^{-3}$

$$X(z)=\frac{1+2z^{-1}+z^{-2}}{1-z^{-1}+0.3561z^{-2}}=1+3z^{-1}+3.6439z^{-2}+2.57z^{-3}$$

$$x(n)=\{1, 3, 3.6439, 2.57, -\}$$

→ ② Recursively method.

$$X(z)=\frac{1+2z^{-1}+z^{-2}}{1-z^{-1}+0.3561z^{-2}}=\frac{q_0+q_1z^{-1}+q_2z^{-2}}{b_0+b_1z^{-1}+b_2z^{-2}}$$

$q_0=1$ $q_1=2$ $q_2=1$

$b_0=1$ $b_1=-1$ $b_2=0.3561$

$$x(0)=\frac{q_0}{b_0}=1$$

$$x(n)=\left[a_n-\sum_{i=1}^{n} x(n-i)b_i\right]/b_0$$

at $n=1$

$$x(1)=q_1-\sum_{i=1}^{1} x(1-i)b_i=q_1-x(0)b_1=2-1(-1)=3$$

at $n=2$

$$x(2)=q_2-\sum_{i=1}^{2} x(2-i)b_i=q_2-[x(1)b_1+x(0)b_2]=3.6439$$

at $n=3$ → $q_3=$ Zero $\quad x(3)=q_3-\sum_{i=1}^{3} x(3-i)b_i$

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② Partial Fraction

$$X(z) = \frac{z^{-1}}{1 - 0.25z^{-1} - 0.375z^{-2}}$$

□ Power of $z \rightarrow$ (+Ve)

$$X(z) = \frac{z^{-1}}{1 - 0.25z^{-1} - 0.375z^{-2}} \cdot \frac{z^2}{z^2} = \frac{z}{z^2 - 0.25z - 0.375}$$

[2] $$\frac{X(z)}{z} = \frac{1}{z^2 - 0.25z - 0.375}$$

$$\frac{X(z)}{z} = \frac{1}{(z+0.5)(z-0.75)} = \frac{A}{z+0.5} + \frac{B}{z-0.75}$$

$$A\Big|_{z=-0.5} = \frac{X(z)}{z}(z+0.5) = \frac{1}{z-0.75} = \frac{1}{-1.25} = \frac{-4}{5}$$

$$B\Big|_{z=0.75} = \frac{X(z)}{z}(z-0.75) = +\frac{4}{5}$$

$$\frac{X(z)}{z} = \frac{-415}{z+0.5} + \frac{415}{z-0.75}$$

$$X(z) = \frac{-415z}{z+0.5} + \frac{415z}{z-0.75}$$

$X(z)$	$x(n)$
$\frac{kz}{z-\alpha}$	$k\alpha^n$
$\frac{k\alpha z}{(z-\alpha)^2}$	$kn\alpha^n$
k	$k\delta(n)$
→ given in Exam ←	

$$x(n) = -\frac{4}{5}(-0.5)^n + \frac{4}{5}(0.75)^n$$

$$x(0) = -\frac{4}{5} + \frac{4}{5} = \text{Zero}$$

$$x(1) = -\frac{4}{5}(-0.5) + \frac{4}{5}(0.75) = +\frac{4}{5} * \frac{1}{2} + \frac{4}{5} \times \frac{3}{4} = \frac{4}{10} + \frac{6}{10} = 1$$

$x(2) =$

$x(3) =$

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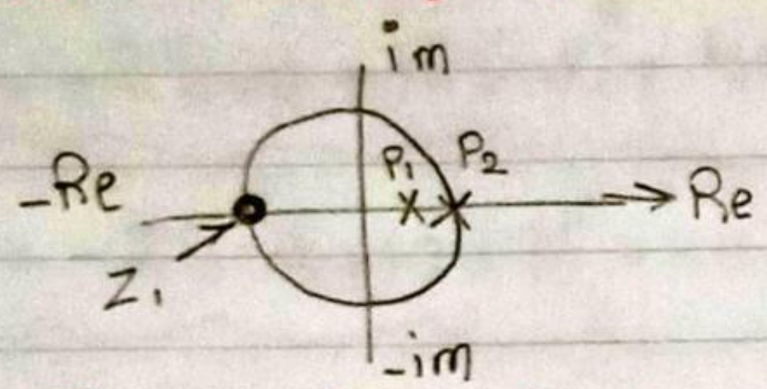
① A digital system has the following Transfer Function. H(z)

$$H(z) = \frac{(Z+1)}{(Z-0.5)(Z-1)^2}$$

ⓐ Plot its Pole-Zero diagram

Zero $Z+1 \longrightarrow Z=1 \rightarrow Z=-1$

Poles $(Z-0.5)(Z-1)^2 \longrightarrow Z=0.5$ ($Z=1$ 2nd order



ⓑ Find The Region of Convergence

→ Roc = All Z except {0.5, 1}

ⓒ Discuss The system stability

→ is marginally stable (Poles on unit circle)

ⓓ using Partial Fraction method Find "4" values of h(n)

solution

$$\frac{H(z)}{Z} = \frac{(Z+1)}{Z(Z-0.5)(Z-1)^2}$$

$$= \frac{A}{Z} + \frac{B}{(Z-0.5)} + \frac{C}{(Z-1)2} + \frac{D}{(Z-1)}$$

$$A\Big|_{Z=0} = \frac{(Z+1)}{(Z-0.5)(Z-1)^2} = \frac{+1}{-0.5*1} = \boxed{+2} \quad (\quad B\Big|_{Z=0.5} = \frac{(Z+1)}{Z(Z-1)^2} = \frac{+1.5}{0.5*0.25} = +12$$

$$C\Big|_{Z=1} = \frac{Z+1}{Z(Z-0.5)} = \frac{2}{0.5} = 4$$

$$D\Big|_{Z=1} = \frac{d}{dz}\frac{(Z+1)}{(Z^2-0.5Z)} = \frac{(Z^2-0.5Z) - (2Z-0.5)(Z+1)}{(Z^2-0.5Z)^2} = \frac{0.5-(1.5X2)}{0.25} = \frac{-2.5}{0.25} = -10$$

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$$\frac{H(z)}{z} = \frac{-2}{z} + \frac{12}{z-0.5} + \frac{4}{(z-1)^2} + \frac{-10}{(z-1)}$$

$$H(z) = -2 + \frac{12z}{z-0.5} + \frac{4z}{(z-1)^2} + \frac{-10z}{(z-1)}$$

$$h(n) = -2\delta(n) + 12(0.5)^n + 4n + -10$$

$$h(n) = -2\delta(n) + 12(0.5)^n + 4n - 10$$

$h(0) = ✓$ $h(1) = ✓$ $h(2) = ✓$ $h(3) = ✓$

$\delta(n)$ → $\delta(0) = 1$

$\delta(n)$ → $\delta(n \neq 0) = Zero$

$h(0) = -2\delta(0) + 12 - 10 = Zero$

$h(1) = -2\delta(1) + 12 \times 0.5 + 4 - 10 = 6 + 4 - 10 = Zero$

$h(2) = -2\delta(2) + 12 * 0.25 + 8 - 10 = 3 + 8 - 10 = 1$

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Difference eQuation

:- is The Relation beTween inPut and outPut in Time domain

$$y(n) = F\{x(n), x(n-1), y(n-m)\}$$

* From The ProPerTies of Z-Transform

  * Delay

$$x(n) \longrightarrow X(z)$$

$$x(n-1) \longrightarrow X(z)z^{-1}$$

→ Find The Difference eQuation

$$H(z) = \frac{1-z^{-1}}{1+0.5z^{-1}}$$

$H(z) \Rightarrow$ is The Transfer Function $= \frac{y(z)}{x(z)}$

$$\frac{y(z)}{x(z)} = \frac{1-z^{-1}}{1+0.5z^{-1}} \qquad y(z)\cdot(1+0.5z^{-1}) = x(z)(1-z^{-1})$$

$$y(z) + 0.5\,y(z)z^{-1} = X(z) - X(z)z^{-1}$$

take iZT For both side

$$y(n) + 0.5\,y(n-1) = x(n) - x(n-1)$$

$$\boxed{y(n) = x(n) - x(n-1) - 0.5\,y(n-1)}$$ Difference eQuation

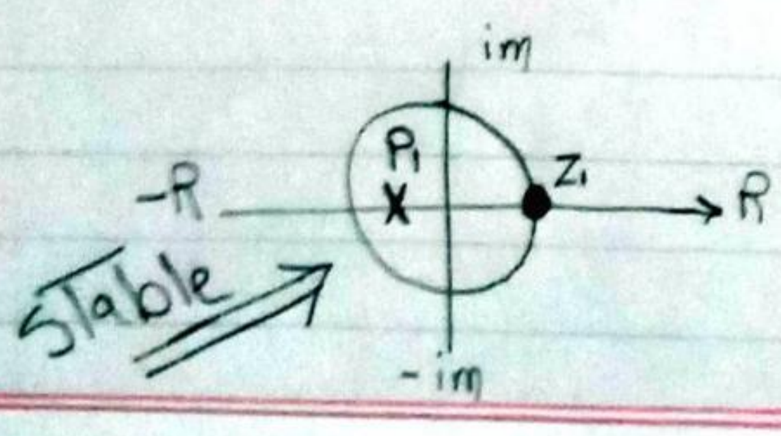
* Draw The Pole-Zero diagram

$$H(z) = \frac{1-z^{-1}}{1+0.5z^{-1}} = \frac{z-1}{z+0.5} \longrightarrow \left(\frac{z}{z}\right) \rightarrow +ve$$

$$H(z) = \frac{\text{Zeros}(z)}{\text{Poles}(z)}$$

Zeros $z-1 \rightarrow z=1$

Poles $z+0.5 \rightarrow z=-0.5$



Stability →
- All Poles inside unit circle → Stable
- outside → unstable
- on → marginally

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